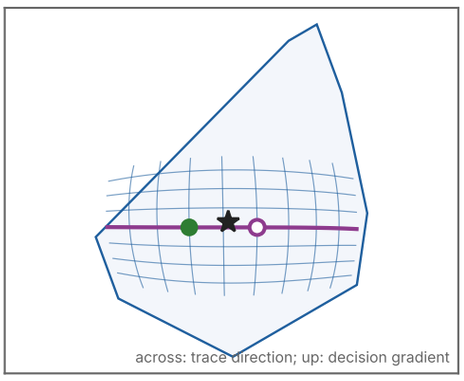
Study 26 at five levels: what each level keeps, and what each map loses

The purple signal is followed through every level: a table on the foreign context {M,R,Y}, refitted at step 3 and later retracted.

observed tables inferred laws selected states the {M,R,Y} history coalition decision threshold twist

1. Toric (Kahler) layer: the fibre in its log-linear chart

computed; a 2-D section of the 9-dimensional fibre, chart lines projected onto it



Each interior law has a log-linear (trace) coordinate, a bijection onto the fibre's positive laws (Thm 146(ii)).

green: maximum entropy, trace 0, the Birch point of the log-linear model (Prop 1(3)); a selected state

purple line: the chart line along the {M,R,Y} trace; hollow circle: history B's end law, TV 0.089 away

star: the leximin law, a different selection: the limit of the soft-min family (Prop 149(ii), Thm 27), not a chart point

torus phases sit over each interior point; not drawn

map 1 to 2: expectation

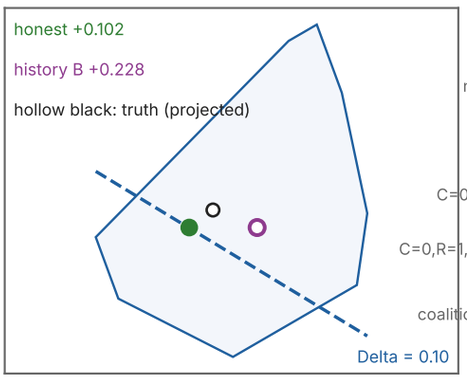
Take expectations E_P[u] of decision functionals.

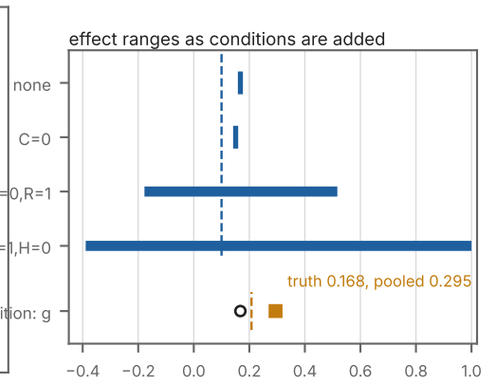
keeps: decision values, identification widths (support function of the fibre)

loses: the chart coordinate: a value does not say which history reached it

2. Probability and decision layer

computed; same section in probability coordinates, with the decision hyperplane; identification widths





map 2 to 3: restriction

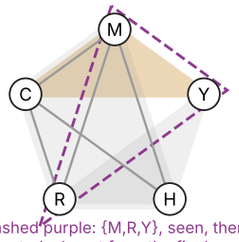
Restrict to contexts: P → (pi_C P) for each table.

keeps: local tables and their supports; compatibility (is there any law?)

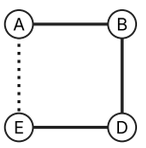
loses: the joint law: which law, and so the decision; the coalition moves a point that restriction still finds compatible

3. Contextuality and sheaf layer

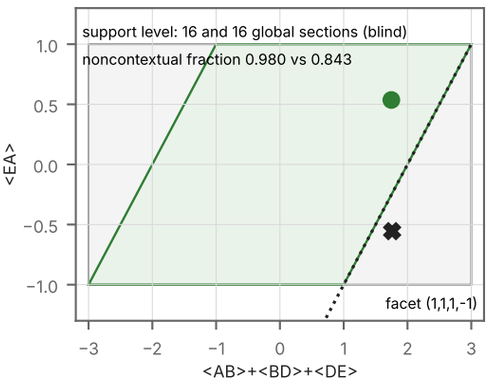
left: observed covers (drawn from the design); right: the cycle's correlations (computed)



dashed purple: {M,R,Y}, seen, then retracted; absent from the final cover



cycle: (E,A) coded differently (the twist)



map 3 to 4: add time

Record the sequence of tables seen, and the updater's state after each.

keeps: the order of revisions; which contexts were ever seen

loses: nothing about the final tables, but the final tables alone lose the history (two histories, same tables, different laws)

4. History and transport layer

computed (Study 26), except the variable-changing row, which is the example of Prop 142

reorder the same tables

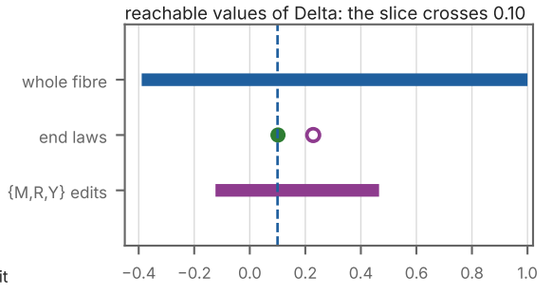
TV 2e-16: path independent (Thm 146(iii))

retract a foreign table {M,R,Y}

TV 0.089; trace in H_D + F_MRY (6e-15), not in H_D + F_MHY (1.6)

change the variables (merge, split)

honest loss H = 0.42; outcome-dependent split launders +-0.59 (Prop 142(iv))



map 4 to 5: encoding E (ours)

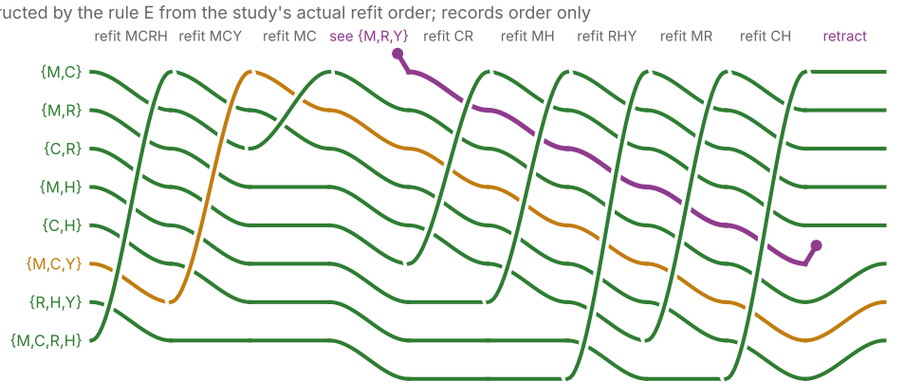
Contexts become strands; a refit moves its strand to the top, over the strands it passes; a table seen then retracted is a strand born and capped.

keeps: the order of interactions (a braid word)

loses: all probability values and the size of the trace: a braid invariant does not determine the law

5. Braid layer: an encoding of history B's refit order

constructed by the rule E from the study's actual refit order; records order only



what the braid keeps

History A is the same picture without the purple strand. The difference is a strand born and capped: the encoding of a newcomer (cf. Def. 133), whose effect on the law the braid does not record.

Co-revision of two sources would be a full twist (Cor. 130); the coalition sits inside one context, so this encoding does not show it.